

WS281x

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Table of contents

Contents

1	How to use	2
1.1	Required .ioc configuration (timer)	2
1.2	Required .ioc configuration (SPI)	2
1.3	Adding library	2
1.4	Configuration file	3
1.5	Initialization	3
1.6	Issues	4
2	Code description	4

1 How to use

1.1 Required .ioc configuration (timer)

Make sure you have these set in your .ioc file:

- In project manager tick *Generate peripheral initialization as pair of '.c/.h' files per peripheral*
- Choose timer and channel which can use DMA
- Timer must be in mode *PWM Generation CHx* where x is number of chosen channel
- Set frequency to 800kHz (1.25 μ s period) on your timer using *Counter Period*. It's not recommended to have prescaler active.
- Add DMA request in DMA settings, choose direction to *Memory to Peripheral*, make sure data width is *Half Word* in Peripheral and Memory. Mode should be *Normal*
- if you use inverting logic level shifter, switch *CH Polarity* to Low

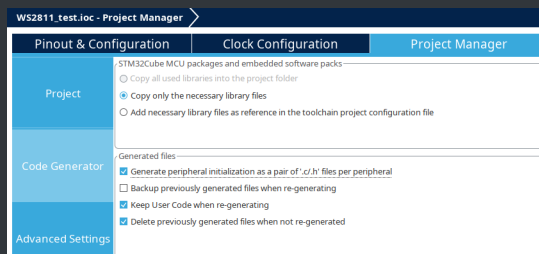


Figure 1: How to generate separated files

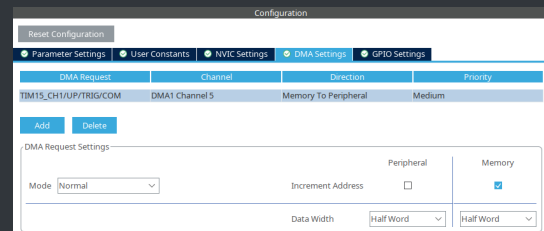


Figure 2: Sample DMA configuration

1.2 Required .ioc configuration (SPI)

Make sure you have these set in your .ioc file:

- In project manager tick *Generate peripheral initialization as pair of '.c/.h' files per peripheral*
- Choose your SPI and then mode: *Half-Duplex Master*
- Set *Data Size* to 6 Bits (if not possible then set 8 Bits), MSB First
- Set your *Prescaler* to value which gives *6-7MBits/s Baud Rate*, you may need to switch clock configuration

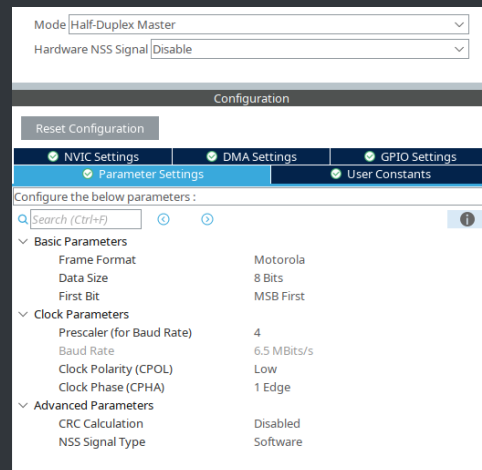


Figure 3: SPI configuration

1.3 Adding library

Just follow these steps:

1. In command line (can be git bash or linux terminal) type:

```
1 git init
2 git submodule add https://github.com/MaciejZb66/ws281x.git
```

2. Refresh project by clicking F5
3. Add library path to build
 - 3.1. Left click on project and choose properties
 - 3.2. On left side choose *C/C++ General*, then *Paths and Symbols* and *includes*
 - 3.3. Click on add button and find folder **ws281x**
 - 3.4. In *Source Location* (still *C/C++ General* and *Paths and Symbols*) add folder **ws281x**
4. In file *main.h* add this line:

```
1 #include "led.h"
```

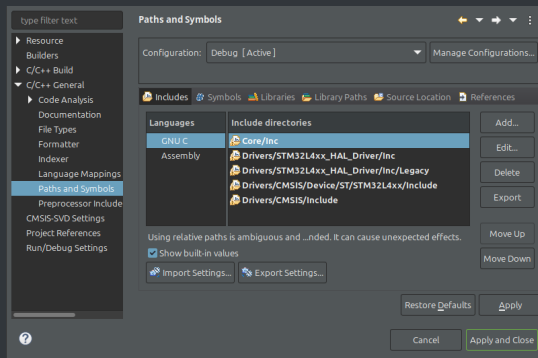


Figure 4: Add path to folder in includes

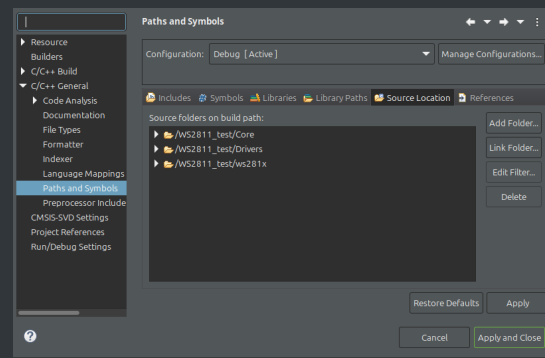


Figure 5: Add to source location

1.4 Configuration file

This library is default configured in *sample_config.h* where we have:

```
1 #define STM_FAMILY_L4
2 #define WS_TIM_15
3 #define WS_CHANNEL_1
4 #define user_leds 24
```

To redefine configuration create your own configuration file. There must be defined:

- using timer or SPI (can be both)
- STM family
- number of outputs
- maximum number of LEDs (WS drivers connected to LEDs)
- `#define WS_CONFIG //required to exclude sample_config.h`

1.5 Initialization

To initialize your chosen timer use function:

```
1 void WS281x_init_TIM(WS281x_data* led, TIM_HandleTypeDef* htim, uint32_t t_channel, uint16_t led_number)
```

To initialize your chosen SPI use function:

```
1 void WS281x_init_SPI(WS281x_data* led, SPI_HandleTypeDef* hspi, uint16_t led_number)
```

and give all required info. *WS281x_data* is named LED and it's created as you define number of outputs. In both cases it's required to initialize once per output.

1.6 Issues

STM32F4 requires 8BIT SPI data size and doesn't work properly. It will be fixed in next updates.

2 Code description

to be described later